

CO4-AT2-LAB PRACTICE

1.JAVA APPLET LIFECYCLE :-

JAVA CODE:

```
import java.applet.Applet;
import java.awt.Graphics;

public class LifeCycle extends Applet {

    public void init() {

        System.out.println("init");

    }

    public void start() {

        System.out.println("start");

    }

    public void paint(Graphics g) {

        g.drawString("Applet Life Cycle ", 100, 100);

        System.out.println("paint");

    }

    public void stop() {

        System.out.println("stop");

    }

    public void destroy() {

        System.out.println("destroy");

    }

}
```

HTML CODE:

```
<html>

<body>

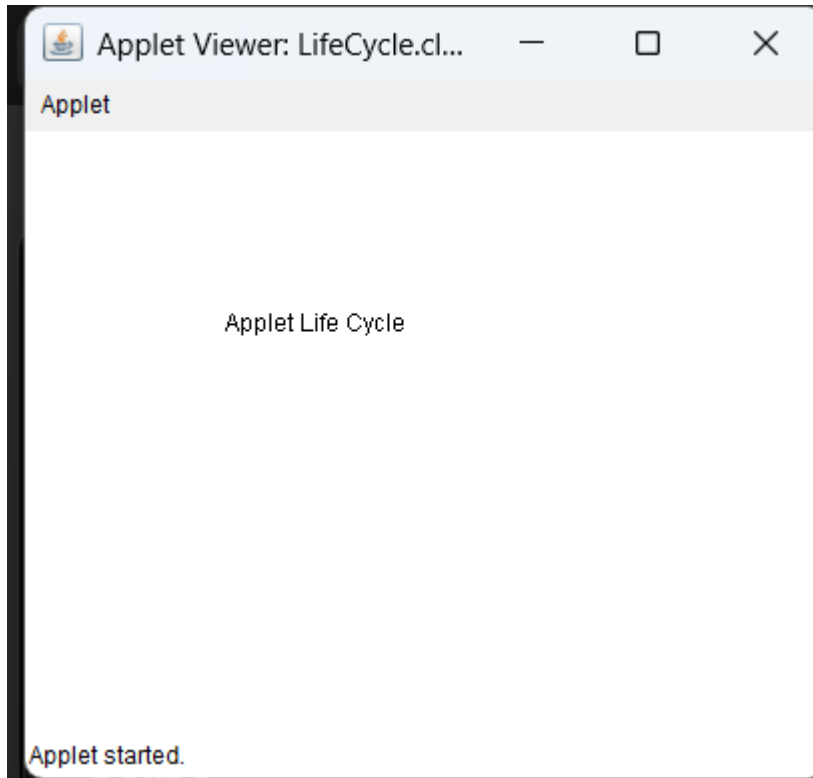
<applet code="LifeCycle.class" width="400" height="300">
```

</applet>

</body>

</html>

OUTPUT:



```
C:\Windows\System32\cmd.e  x  +  v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HEMAA\OneDrive\Pictures\JAVA\Applet>"C:\Program Files\Java\jdk1.8.0_202\bin\appletviewer.exe" LifeCycle.html
init
start
paint
stop
destroy

C:\Users\HEMAA\OneDrive\Pictures\JAVA\Applet>"C:\Program Files\Java\jdk1.8.0_202\bin\appletviewer.exe" LifeCycle.html
init
start
paint
stop
destroy

C:\Users\HEMAA\OneDrive\Pictures\JAVA\Applet>|
```

2.JAVA APPLET SHAPES : -

JAVA CODE:

```
import java.applet.Applet;
import java.awt.*;

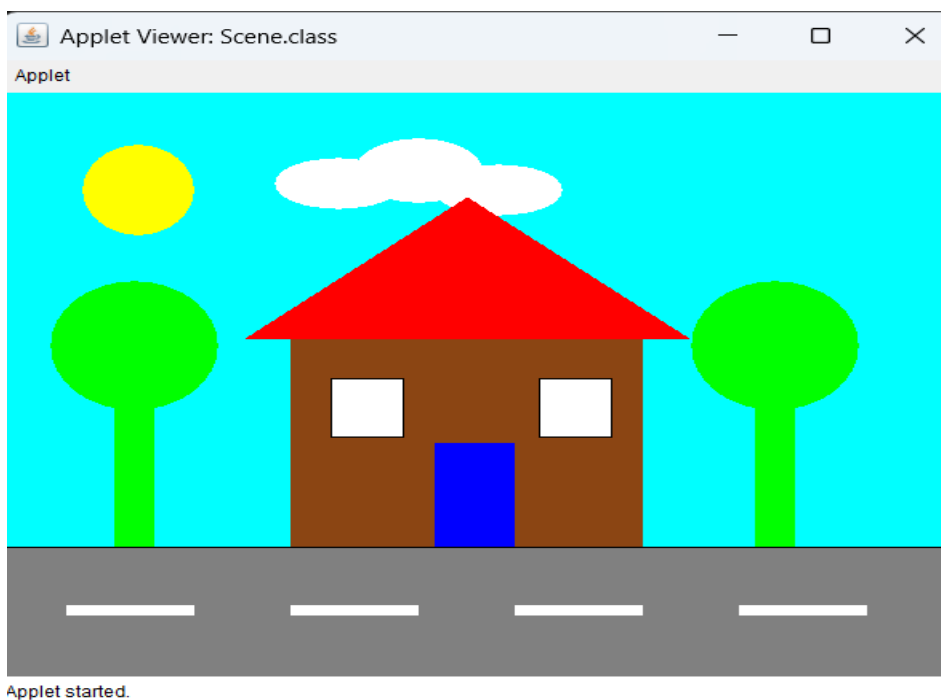
public class Scene extends Applet {
    public void paint(Graphics g) {
        setBackground(Color.CYAN);
        g.setColor(Color.YELLOW);
        g.fillOval(50, 40, 70, 70);
        g.setColor(Color.WHITE);
        g.fillOval(170, 50, 80, 40);
        g.fillOval(220, 35, 80, 50);
        g.fillOval(270, 55, 80, 40);
        g.setColor(Color.GRAY);
        g.fillRect(0, 350, 600, 100);
        g.setColor(Color.WHITE);
        g.fillRect(40, 395, 80, 8);
        g.fillRect(180, 395, 80, 8);
        g.fillRect(320, 395, 80, 8);
        g.fillRect(460, 395, 80, 8);
        g.setColor(new Color(139, 69, 19));
        g.fillRect(180, 190, 220, 160);
        g.setColor(Color.RED);
        Polygon roof = new Polygon();
        roof.addPoint(150, 190);
        roof.addPoint(290, 80);
        roof.addPoint(430, 190);
        g.fillPolygon(roof);
        g.setColor(Color.BLUE);
```

```

    g.fillRect(270, 270, 50, 80);
    g.setColor(Color.WHITE);
    g.fillRect(205, 220, 45, 45);
    g.fillRect(335, 220, 45, 45);
    g.setColor(Color.BLACK);
    g.drawRect(205, 220, 45, 45);
    g.drawRect(335, 220, 45, 45);
    g.setColor(Color.GREEN);
    g.fillRect(70, 210, 25, 140);
    g.fillOval(30, 145, 105, 100);
    g.fillRect(470, 210, 25, 140);
    g.fillOval(430, 145, 105, 100);
    g.setColor(Color.BLACK);
    g.drawLine(0, 350, 600, 350);
}
}

```

OUTPUT:



3.JAVA APPLET CALCULATOR :-

JAVA CODE:

```
import java.applet.Applet;
import java.awt.Graphics;

public class ParameterCalculator extends Applet {

    double num1, num2, result;

    String operator;

    public void init() {

        num1 = Double.parseDouble(getParameter("num1"));
        num2 = Double.parseDouble(getParameter("num2"));
        operator = getParameter("operator");
        if (operator.equals("+"))
            result = num1 + num2;
        else if (operator.equals("-"))
            result = num1 - num2;
        else if (operator.equals("*"))
            result = num1 * num2;
        else if (operator.equals("/"))
            result = num1 / num2;
    }

    public void paint(Graphics g) {

        g.drawString("Number 1: " + num1, 50, 60);
        g.drawString("Number 2: " + num2, 50, 90);
        g.drawString("Operator: " + operator, 50, 120);
        g.drawString("Result: " + result, 50, 150);
    }
}
```

HTML CODE:

```
<html>
```

```
<body>

<applet code="ParameterCalculator.class" width="400" height="250">

<param name="num1" value="20">

<param name="num2" value="5">

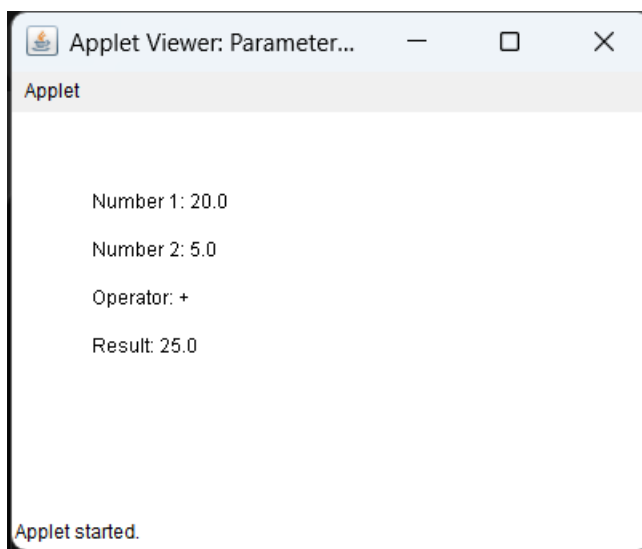
<param name="operator" value="+">

</applet>

</body>

</html>
```

OUTPUT:



4.STUDENT PROFILE:-

JAVA CODE:

```
import java.awt.*;

import java.awt.event.*;

public class StudentProfile extends Frame implements ActionListener {

    TextField name, regno;

    Checkbox male, female;

    CheckboxGroup gender;

    Choice department, year;

    List skills;

    TextArea output;
```

Button generate;

```
public StudentProfile() {  
    setTitle("Student Profile");  
    setSize(650, 650);  
    setLayout(new BorderLayout(10, 10));  
    setBackground(Color.WHITE);  
    Label heading = new Label("STUDENT PROFILE", Label.CENTER);  
    heading.setFont(new Font("Arial", Font.BOLD, 24));  
    heading.setForeground(Color.BLUE);  
    add(heading, BorderLayout.NORTH);  
    Panel form = new Panel(new GridLayout(6, 2, 10, 10));  
    form.setBackground(new Color(220, 240, 255));  
    form.add(new Label("Name:"));  
    name = new TextField();  
    form.add(name);  
    form.add(new Label("Register Number:"));  
    regno = new TextField();  
    form.add(regno);  
    form.add(new Label("Gender:"));  
    Panel genderPanel = new Panel();  
    gender = new CheckboxGroup();  
    male = new Checkbox("Male", gender, false);  
    female = new Checkbox("Female", gender, false);  
    genderPanel.add(male);  
    genderPanel.add(female);  
    form.add(genderPanel);  
    form.add(new Label("Department:"));  
    department = new Choice();  
    department.add("CSE");  
}
```

```
department.add("IT");
department.add("ECE");
department.add("EEE");
department.add("MECH");
form.add(department);
form.add(new Label("Year:"));
year = new Choice();
year.add("1st Year");
year.add("2nd Year");
year.add("3rd Year");
year.add("4th Year");
form.add(year);
form.add(new Label("Skills / Programming Languages:"));
skills = new List(4, true);
skills.add("Java");
skills.add("Python");
skills.add("C");
skills.add("C++");
skills.add("HTML");
skills.add("SQL");
form.add(skills);
generate = new Button("Generate Report");
generate.setBackground(Color.GREEN);
generate.setForeground(Color.WHITE);
generate.addActionListener(this);
Panel center = new Panel(new BorderLayout(10, 10));
center.add(form, BorderLayout.CENTER);
center.add(generate, BorderLayout.SOUTH);
add(center, BorderLayout.CENTER);
```

```

output = new TextArea();
output.setEditable(false);
output.setBackground(new Color(245, 235, 255));
add(output, BorderLayout.SOUTH);
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        System.exit(0);
    }
});
setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    String selectedGender = "";
    if (male.getState())
        selectedGender = "Male";
    else if (female.getState())
        selectedGender = "Female";
    String[] selectedSkills = skills.getSelectedItems();
    String skillText = "";
    for (int i = 0; i < selectedSkills.length; i++) {
        skillText += selectedSkills[i] + " ";
    }
    output.setText(
        "STUDENT PROFILE\n\n" +
        "Name: " + name.getText() + "\n" +
        "Register Number: " + regno.getText() + "\n" +
        "Gender: " + selectedGender + "\n" +
        "Department: " + department.getSelectedItem() + "\n" +
        "Year: " + year.getSelectedItem() + "\n" +

```

```

        "Skills: " + skillText
    );
}

public static void main(String[] args) {
    new StudentProfile();
}
}

```

OUTPUT:

STUDENT PROFILE

Name: Hema

Register Number: 192524467

Gender: ☐ Male ☒ Female

Department: CSE

Year: 2nd Year

Skills / Programming Languages: Java, Python, C

Generate Report

STUDENT PROFILE

Name: Hema
 Register Number: 192524467
 Gender: Female
 Department: CSE
 Year: 2nd Year
 Skills: Java Python C

5.GRID LAYOUT :-

JAVA CODE:

```

import java.awt.*;
import java.awt.event.*;

```

```

public class GridCalculator extends Frame implements ActionListener {
    TextField display;
    double num1, num2, result;
    char operator;
    public GridCalculator() {
        setTitle("Grid Calculator");
        setSize(400, 500);
        setLayout(new BorderLayout(10, 10));
        display = new TextField();
        display.setEditable(false);
        display.setFont(new Font("Arial", Font.BOLD, 24));
        add(display, BorderLayout.NORTH);
        Panel panel = new Panel();
        panel.setLayout(new GridLayout(4, 4, 5, 5));
        String[] buttons = {
            "7", "8", "9", "+",
            "4", "5", "6", "-",
            "1", "2", "3", "*",
            "0", "=", "C", "/"
        };
        for (int i = 0; i < buttons.length; i++) {
            Button b = new Button(buttons[i]);
            b.setFont(new Font("Arial", Font.BOLD, 20));
            b.addActionListener(this);
            panel.add(b);
        }
        add(panel, BorderLayout.CENTER);
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {

```

```

        System.exit(0);
    }
});
setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    String s = e.getActionCommand();
    if (s.equals("C")) {
        display.setText("");
        num1 = 0;
        num2 = 0;
        result = 0;
    }
    else if (s.equals("+") || s.equals("-") ||
             s.equals("*") || s.equals("/")) {
        if (!display.getText().equals("")) {
            num1 = Double.parseDouble(display.getText());
            operator = s.charAt(0);
            display.setText("");
        }
    }
    else if (s.equals("=")) {
        if (!display.getText().equals("")) {
            num2 = Double.parseDouble(display.getText());
            if (operator == '+')
                result = num1 + num2;
            else if (operator == '-')
                result = num1 - num2;
            else if (operator == '*')

```

```

        result = num1 * num2;
    else if (operator == '/')
        result = num1 / num2;
    display.setText(String.valueOf(result));
}
}
else {
    display.setText(display.getText() + s);
}
}

public static void main(String[] args) {
    new GridCalculator();
}
}

```

OUTPUT:



6.GRIDBAG LAYOUT :-

JAVA CODE:

```
import java.awt.*;
import java.awt.event.*;

public class StudentRegistration extends Frame implements ActionListener {

    TextField name, regno, email;

    Choice department, year;

    Checkbox male, female;

    CheckboxGroup gender;

    Button submit, reset;

    TextArea output;

    public StudentRegistration() {

        setTitle("Student Registration Form");

        setSize(650, 650);

        setLayout(new GridBagLayout());

        setBackground(Color.WHITE);

        GridBagConstraints gbc = new GridBagConstraints();

        gbc.insets = new Insets(8, 8, 8, 8);

        gbc.fill = GridBagConstraints.HORIZONTAL;

        Label heading = new Label("STUDENT REGISTRATION FORM", Label.CENTER);

        heading.setFont(new Font("Arial", Font.BOLD, 24));

        heading.setForeground(Color.BLUE);

        gbc.gridx = 0;

        gbc.gridy = 0;

        gbc.gridwidth = 2;

        add(heading, gbc);

        gbc.gridwidth = 1;

        gbc.gridx = 0;

        gbc.gridy = 1;
```

```
add(new Label("Name:"), gbc);
name = new TextField(20);
gbc.gridx = 1;
add(name, gbc);
gbc.gridx = 0;
gbc.gridy = 2;
add(new Label("Register Number:"), gbc);
regno = new TextField(20);
gbc.gridx = 1;
add(regno, gbc);
gbc.gridx = 0;
gbc.gridy = 3;
add(new Label("Department:"), gbc);
department = new Choice();
department.add("CSE");
department.add("IT");
department.add("ECE");
department.add("EEE");
department.add("MECH");
gbc.gridx = 1;
add(department, gbc);
gbc.gridx = 0;
gbc.gridy = 4;
add(new Label("Email:"), gbc);
email = new TextField(20);
gbc.gridx = 1;
add(email, gbc);
gbc.gridx = 0;
gbc.gridy = 5;
```

```
add(new Label("Gender:"), gbc);
Panel genderPanel = new Panel();
gender = new CheckboxGroup();
male = new Checkbox("Male", gender, false);
female = new Checkbox("Female", gender, false);
genderPanel.add(male);
genderPanel.add(female);
gbc.gridx = 1;
add(genderPanel, gbc);
gbc.gridx = 0;
gbc.gridy = 6;
add(new Label("Year:"), gbc);
year = new Choice();
year.add("1st Year");
year.add("2nd Year");
year.add("3rd Year");
year.add("4th Year");
gbc.gridx = 1;
add(year, gbc);
submit = new Button("Submit");
submit.setBackground(Color.GREEN);
submit.setForeground(Color.WHITE);
submit.addActionListener(this);
gbc.gridx = 0;
gbc.gridy = 7;
add(submit, gbc);
reset = new Button("Reset");
reset.setBackground(Color.RED);
reset.setForeground(Color.WHITE);
```

```

reset.addActionListener(this);

gbc.gridx = 1;
add(reset, gbc);

output = new TextArea(5, 30);
output.setEditable(false);
output.setBackground(new Color(235, 245, 255));

gbc.gridx = 0;
gbc.gridy = 8;
gbc.gridwidth = 2;
add(output, gbc);

addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        System.exit(0);
    }
});

setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    if (e.getSource() == submit) {
        String selectedGender = "";

        if (male.getState())
            selectedGender = "Male";
        else if (female.getState())
            selectedGender = "Female";

        output.setText(
            "STUDENT REGISTRATION DETAILS\n\n" +
            "Name: " + name.getText() + "\n" +
            "Register Number: " + regno.getText() + "\n" +
            "Department: " + department.getSelectedItem() + "\n" +

```

```

        "Email: " + email.getText() + "\n" +
        "Gender: " + selectedGender + "\n" +
        "Year: " + year.getSelectedItem()
    );
}
if (e.getSource() == reset) {
    name.setText("");
    regno.setText("");
    email.setText("");
    output.setText("");
    department.select(0);
    year.select(0);
    male.setState(false);
    female.setState(false);
}
}
public static void main(String[] args) {
    new StudentRegistration();
}
}

```

OUTPUT:

STUDENT REGISTRATION FORM

Name:

Register Number:

Department:

Email:

Gender: ☐ Male ☒ Female

Year:

STUDENT REGISTRATION DETAILS

Name: Hema
Register Number: 192524467
Department: CSE
Email: hema@gmail.com

STUDENT REGISTRATION FORM

Name:

Register Number:

Department:

Email:

Gender: ☐ Male ☒ Female

Year:

Name: Hema
Register Number: 192524467
Department: CSE
Email: hema@gmail.com
Gender: Female
Year: 2nd Year

7.BORDER LAYOUT:

JAVA CODE:

```
import java.awt.*;
import java.awt.event.*;

public class BorderLayoutDemo extends Frame implements ActionListener {

    TextArea display;

    Button home, profile, settings;

    Button option1, option2;

    Button submit, clear;

    public BorderLayoutDemo() {

        setTitle("BorderLayout Application");

        setSize(700, 500);

        setLayout(new BorderLayout(10, 10));

        Label header = new Label("STUDENT INFORMATION SYSTEM", Label.CENTER);
        header.setFont(new Font("Arial", Font.BOLD, 22));
        header.setForeground(Color.BLUE);
        add(header, BorderLayout.NORTH);

        Panel menu = new Panel(new GridLayout(3, 1, 5, 5));

        home = new Button("Home");
        profile = new Button("Profile");
        settings = new Button("Settings");

        menu.add(home);
        menu.add(profile);
        menu.add(settings);

        home.addActionListener(this);
        profile.addActionListener(this);
        settings.addActionListener(this);

        add(menu, BorderLayout.WEST);

        Panel options = new Panel(new GridLayout(2, 1, 5, 5));
```

```
option1 = new Button("Option 1");
option2 = new Button("Option 2");
options.add(option1);
options.add(option2);
option1.addActionListener(this);
option2.addActionListener(this);
add(options, BorderLayout.EAST);
display = new TextArea();
display.setEditable(false);
display.setText("Welcome to Student Information System");
display.setFont(new Font("Arial", Font.PLAIN, 16));
add(display, BorderLayout.CENTER);
Panel buttons = new Panel();
submit = new Button("Submit");
clear = new Button("Clear");
submit.setBackground(Color.GREEN);
submit.setForeground(Color.WHITE);
clear.setBackground(Color.RED);
clear.setForeground(Color.WHITE);
buttons.add(submit);
buttons.add(clear);
submit.addActionListener(this);
clear.addActionListener(this);
add(buttons, BorderLayout.SOUTH);
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        System.exit(0);
    }
});
```

```

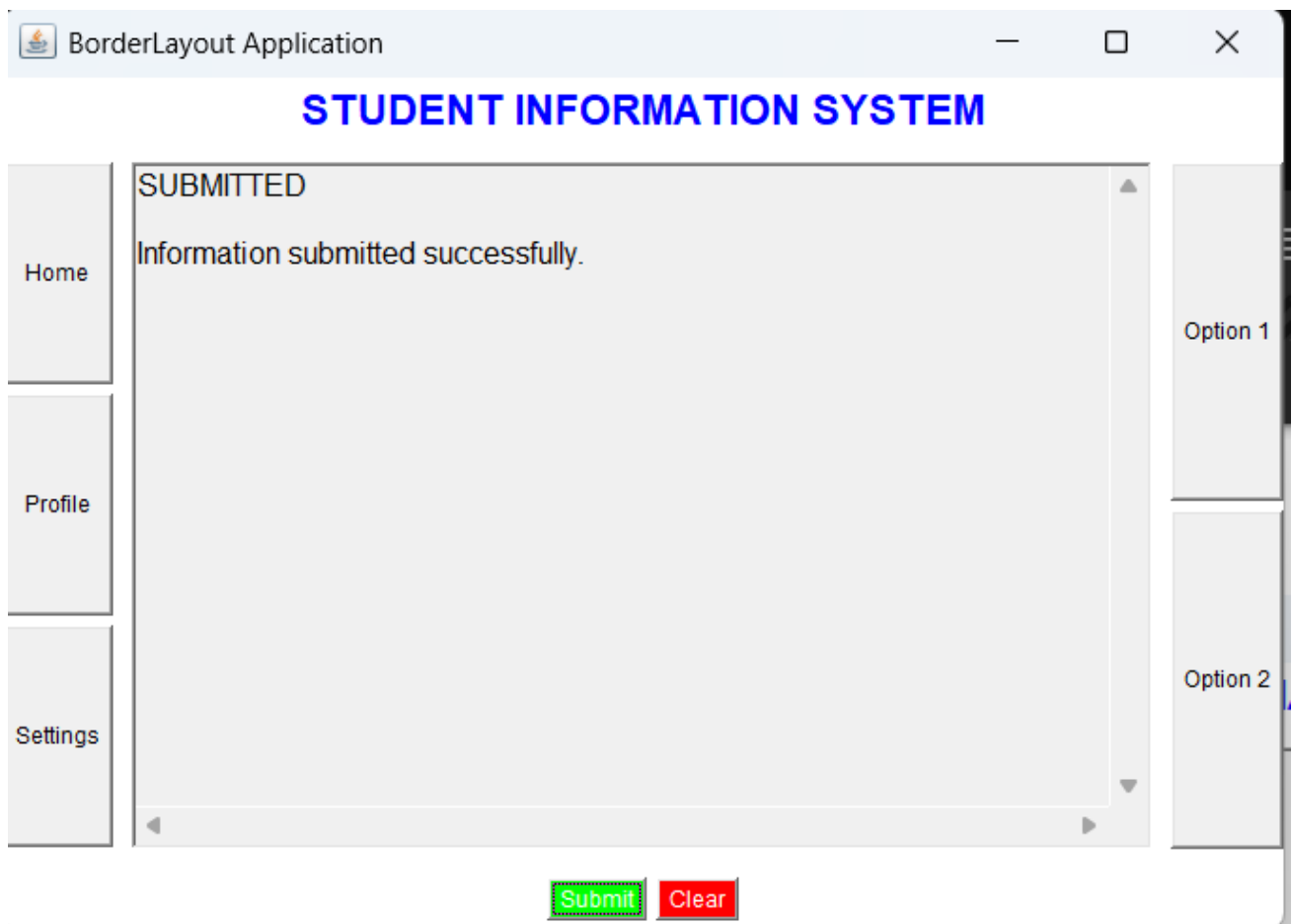
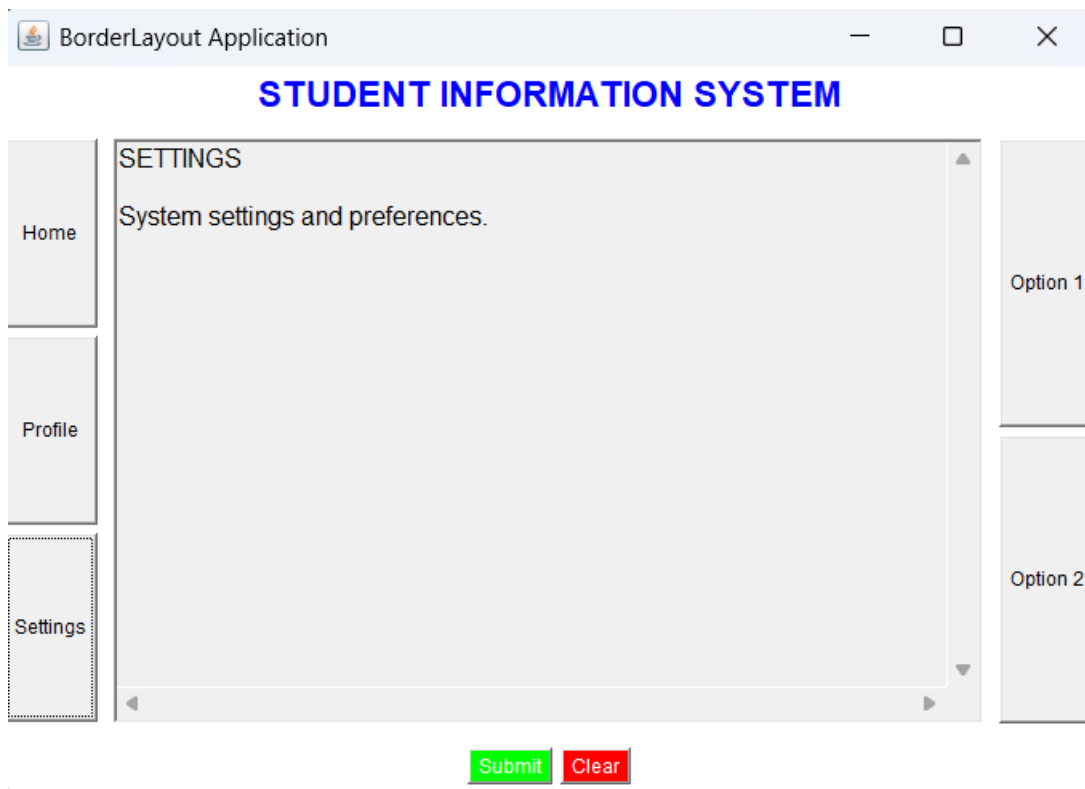
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        String button = e.getActionCommand();
        if (button.equals("Home"))
            display.setText("HOME\n\nWelcome to the Student Information System.");
        else if (button.equals("Profile"))
            display.setText("PROFILE\n\nStudent profile information.");
        else if (button.equals("Settings"))
            display.setText("SETTINGS\n\nSystem settings and preferences.");
        else if (button.equals("Option 1"))
            display.setText("OPTION 1\n\nOption 1 selected.");
        else if (button.equals("Option 2"))
            display.setText("OPTION 2\n\nOption 2 selected.");
        else if (button.equals("Submit"))
            display.setText("SUBMITTED\n\nInformation submitted successfully.");
        else if (button.equals("Clear"))
            display.setText("");
    }

    public static void main(String[] args) {
        new BorderLayoutDemo();
    }
}

```

OUTPUT:



8.CARD LAYOUT:-

JAVA CODE:

```
import java.awt.*;
import java.awt.event.*;

public class CardLayoutDemo extends Frame implements ActionListener {

    CardLayout cardLayout;

    Panel cards;

    Button next, previous, submit;

    TextField name, regno, course, marks;

    Choice year, semester;

    TextArea report;

    public CardLayoutDemo() {

        setTitle("Student Information");

        setSize(600, 450);

        setLayout(new BorderLayout());

        cardLayout = new CardLayout();

        cards = new Panel(cardLayout);

        Panel card1 = new Panel(new GridLayout(3, 2, 10, 10));

        card1.add(new Label("Student Name:"));

        name = new TextField();

        card1.add(name);

        card1.add(new Label("Register Number:"));

        regno = new TextField();

        card1.add(regno);

        card1.add(new Label("Course:"));

        course = new TextField();

        card1.add(course);

        Panel card2 = new Panel(new GridLayout(3, 2, 10, 10));

        card2.add(new Label("Year:"));
```

```
year = new Choice();
year.add("1st Year");
year.add("2nd Year");
year.add("3rd Year");
year.add("4th Year");
card2.add(year);
card2.add(new Label("Semester:"));
semester = new Choice();
semester.add("Semester 1");
semester.add("Semester 2");
semester.add("Semester 3");
semester.add("Semester 4");
semester.add("Semester 5");
semester.add("Semester 6");
semester.add("Semester 7");
semester.add("Semester 8");
card2.add(semester);
card2.add(new Label("Marks:"));
marks = new TextField();
card2.add(marks);

Panel card3 = new Panel(new BorderLayout());
report = new TextArea();
report.setEditable(false);
card3.add(report, BorderLayout.CENTER);
cards.add(card1, "Student Information");
cards.add(card2, "Academic Information");
cards.add(card3, "Confirmation/Report");
add(cards, BorderLayout.CENTER);

Panel buttons = new Panel();
```

```

previous = new Button("Previous");
next = new Button("Next");
submit = new Button("Submit");
buttons.add(previous);
buttons.add(next);
buttons.add(submit);
previous.addActionListener(this);
next.addActionListener(this);
submit.addActionListener(this);
add(buttons, BorderLayout.SOUTH);
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        System.exit(0);
    }
});
setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    String command = e.getActionCommand();
    if (command.equals("Next")) {
        cardLayout.next(cards);
    }
    else if (command.equals("Previous")) {
        cardLayout.previous(cards);
    }
    else if (command.equals("Submit")) {
        report.setText(
            "CONFIRMATION / REPORT\n\n" +
            "Student Name: " + name.getText() + "\n" +

```

```
        "Register Number: " + regno.getText() + "\n" +  
        "Course: " + course.getText() + "\n" +  
        "Year: " + year.getSelectedItem() + "\n" +  
        "Semester: " + semester.getSelectedItem() + "\n" +  
        "Marks: " + marks.getText()  
    );  
    cardLayout.show(cards, "Confirmation/Report");  
}  
}  
public static void main(String[] args) {  
    new CardLayoutDemo();  
}  
}
```

OUTPUT:

Student Information

Year:

2nd Year

Semester:

Semester 4

Marks:

100

Previous

Next

Submit

Student Information

Year:

2nd Year

Semester:

Semester 4

Marks:

100

Previous

Next

Submit



Student Information



CONFIRMATION / REPORT

Student Name: Hema

Register Number: 192524467

Course: Java

Year: 2nd Year

Semester: Semester 4

Marks: 100

Previous

Next

Submit